

set the table and pour me some water

two SO-101 arms . plans validated against physics . 100/100 seeds

# Apparecchiato

Two SO-101 arms set a dinner table from a spoken command.

They hand the plate across when neither arm can reach alone,  
and the hand-off is derived from geometry, not scripted.

AI Infra Summit Hackathon Intel track, Bimanual VLA Manipulation with Multi-Modal Reasoning Online

Gabriele Desimini solo [github.com/dabiH2/apparecchiato](https://github.com/dabiH2/apparecchiato)

# The model decides what. Geometry decides whether it is possible.

One end-to-end policy for a multi-step bimanual task is the obvious answer, and it is the one that fails quietly. A policy that memorises a layout looks excellent on the scene it was tuned on and collapses when the plate moves.

## **A validator sits on the boundary.**

Plans that reference objects not on the table, pick cutlery from a drawer they never opened, place what was never picked, or pin an arm to a target outside its workspace are rejected with a specific reason, before a joint moves.

## **When the model fails, the robot keeps working.**

A deterministic planner takes over and the fallback is recorded in the plan. That boundary is why this runs unattended across randomised seeds.

# Nothing names the object that gets passed.

The scheduler emits a hand-off exactly when the set of arms that can PICK something and the set that can PLACE it do not intersect. On this table that is the plate: it spawns where only the left arm reaches, and its slot is where only the right arm reaches.

**Tested, not asserted.** tests/test\_handoff\_is\_emergent.py moves the plate within reach of its own slot and the hand-off disappears. Same seed, same planner, same code, one fewer step in the plan.

**Being precise about how far the claim goes.** Across a hundred seeds it is always the plate and always left to right. That is the task, not a script, and what the seed varies is where the transfer happens.

# 100% across 100 randomised seeds.

Eleven steps of eleven, every seed. All seven subgoals at 100%. Randomised placement, mass, friction, object scale, lighting and background colour.

## Four qualifications, stated here rather than buried:

1. The VLM's plan is rejected on every seed. The 100% is the fallback's.
2. Those numbers use simulator state, not perception. Perception is measured separately: 7/10 driving two objects, 0/10 driving all five.
3. The Intel figures were measured on an AMD host. No Core Ultra was available.
4. 100/100 is a rate over seeds screened for reachability, at a 45 mm tolerance.

Every number above is produced by the harness and committed under results/.

# OpenVINO, and one finding that transfers.

**Qwen2-VL-2B planner exported to OpenVINO IR at INT4**, OWLv2 detector at INT8, both by hand through `torch.export`, `ov.convert_model` and `nncf.compress_weights`, because optimum-intel has no export for zero-shot object detection.

**bench\_openvino.py sweeps every device against every precision** rather than trusting AUTO, and prints the host CPU so a latency is never quoted without the silicon it came from.

**The CPU plugin silently runs bf16**. A 0.17 mean logit shift looked exactly like a broken export. Pinning `INFERENCE_PRECISION_HINT` to f32 brought it to 2.1e-05. A latency number without its precision is not a measurement.

WHAT WE MEASURED THAT NOBODY ASKED FOR

# The uncomfortable questions, answered first.

**Is the vision half doing anything?** Instruction and scene text held fixed, image varied. No image changes the output. A blank frame changes it again. Another seed's frame gives byte-identical output. So the vision tower contributes 'a photograph of a table', not 'the plate is here'.

**Can the validator repair instead of only rejecting?** It may delete and reorder what the model wrote, and may never add to it, enforced in code. It made no plan executable. It moved the failure from the parser to the validator, where the error names the model's actual mistake.

**Can the copy drift from the code?** Not any more. `audit_claims.py` fails on any retracted claim, anywhere in the repo, the docs or the judging packet.